

Individual Plant & Its Requirements

AgriSpike Crop Management & Cultivation Guidance System

1. TOMATO

Scientific Name: *Solanum lycopersicum* | **Type:** Vegetable | **Family:** Solanaceae

SOIL REQUIREMENTS

- Well-drained soil is preferred.
- Soil should be rich in organic matter.
- Suitable pH: **6.5 – 7.5**.
- Water should not remain stagnant around the roots.

NUTRIENT REQUIREMENTS

- **Nitrogen (N):** Leaf and plant growth
- **Phosphorus (P):** Root and flower development
- **Potassium (K):** Fruit development and quality
- **Fertigation (TNAU):** Recommended dose of **200:250:250 kg/ha N:P:K** for hybrids (fertilizer recommendation, not a sensor target).

IRRIGATION

- Needs regular moisture. Avoid both very dry soil and waterlogging.
- Drip irrigation is suitable for precision irrigation, adjusted according to soil moisture and growth stage.

ENVIRONMENTAL PARAMETERS

Parameter	Requirement
Temperature	Around 20 – 30°C
Humidity	Moderate humidity preferred
Sunlight	Good sunlight required
Soil pH	6.5 – 7.5
Soil Type	Well-drained loamy soil

IMPORTANT PROBLEMS

- Leaf diseases & Fruit diseases
- Insect pests & Nutrient deficiency
- Water stress

AgriSpike Recommendation Example

- Low soil moisture → **Irrigation required**
- Sufficient soil moisture → **No irrigation required**
- Low Nitrogen → **Nitrogen deficiency warning**
- Very high temperature → **Heat-stress warning**

2. CHILLI

Scientific Name: *Capsicum annuum* | **Type:** Vegetable | **Family:** Solanaceae

SOIL REQUIREMENTS

- Well-drained loamy / sandy loam soil is suitable.
- Soil should contain good organic matter.
- Suitable pH: **6.5 – 7.5** (according to TNAU guidance).

NUTRIENT REQUIREMENTS

- **Nitrogen (N):** Plant and leaf growth
- **Phosphorus (P):** Root and flowering
- **Potassium (K):** Fruit development and quality
- **TNAU Recommendation:** Basic dose of **30:60:30 kg/ha N:P:K** for varieties with additional N top dressing; differs for hybrids.

IRRIGATION

- After seedlings are established, irrigation is generally done at weekly intervals (TNAU).
- Adjust during very hot/dry conditions. Drip irrigation can be used for precision water delivery.

ENVIRONMENTAL PARAMETERS

Parameter	Requirement
Temperature	20 – 25°C ideal (fruit set affected at ≥ 37°C)
Humidity	Warm and humid during growth
Sunlight	Good sunlight
Soil pH	6.5 – 7.5
Soil Type	Well-drained loam / sandy loam

PROBLEMS TO MONITOR

- Aphids & Thrips
- Fruit damage & Fungal diseases
- Nutrient deficiency & Excess moisture

AgriSpike Recommendation

- Low moisture → **Irrigation required**
- High moisture → **Stop irrigation**
- High temperature → **Heat warning**
- Low N → **Nitrogen deficiency warning**
- Heavy rain → **Reduce irrigation**

3. BRINJAL (Eggplant)

Scientific Name: *Solanum melongena* | **Type:** Vegetable | **Family:** Solanaceae

SOIL REQUIREMENTS

- Well-drained fertile soil rich in organic matter.
- Suitable pH: **6.5 – 7.5** according to TNAU guidance.
- Avoid excessive water around the root zone.

NUTRIENT REQUIREMENTS

- **Primary Nutrients:** Nitrogen (N), Phosphorus (P), Potassium (K).
- **Soil Health:** Monitor organic matter, general fertility, and deficiencies.

ENVIRONMENTAL PARAMETERS

Parameter	Requirement
Temperature	Around 20 – 30°C favourable
Humidity	Moderate
Sunlight	Good sunlight
Soil pH	6.5 – 7.5
Soil Type	Well-drained fertile soil

PROBLEMS TO MONITOR

- Shoot and fruit borer
- Leaf diseases & Wilt
- Nutrient deficiency & Water stress

IRRIGATION

- After seedlings establish, TNAU recommends weekly irrigation. Drip irrigation can be effectively used.

AgriSpike Recommendation

- Low moisture → **Irrigate**
- Good moisture → **No irrigation**
- Excess moisture → **Warning**
- Low nutrient → **Nutrient warning**
- Pest detected → **Pest alert**

4. ONION

Scientific Name: *Allium cepa* | **Type:** Vegetable | **Family:** Alliaceae

SOIL REQUIREMENTS

- Red loam to black soil with good drainage.
- Small onion performs well at approximately pH 6 – 7.
- Heavy waterlogging should be strictly avoided.

NUTRIENT REQUIREMENTS

- **Important Nutrients:** Nitrogen (N), Phosphorus (P), Potassium (K).
- **TNAU Recommendation:** Basal dose of **30 kg N, 60 kg P, 30 kg K / ha**, plus additional N later in the crop cycle.

ENVIRONMENTAL PARAMETERS

Parameter	Requirement
Temperature	Mild / cool conditions preferred
Humidity	Moderate
Sunlight	Good sunlight
Soil pH	6.0 – 7.0
Soil Type	Well-drained red loam / black soil

PROBLEMS TO MONITOR

- Thrips & Fungal diseases
- Bulb development problems
- Nutrient deficiency & Excess water

IRRIGATION SCHEDULE (TNAU)

- Irrigation at planting → Irrigation on 3rd day → Weekly intervals thereafter.
- **Harvest Rule:** Stop irrigation around 10 days before harvest.

AgriSpike Recommendation

- Low moisture → **Irrigation required**
- Sufficient moisture → **No irrigation**
- Heavy rain → **Stop irrigation**
- Low N → **Nitrogen warning**
- Near harvest → **Reduce / stop irrigation**

5. BANANA

Scientific Name: *Musa spp.* | **Type:** Fruit Crop

SOIL REQUIREMENTS

- Deep, fertile, loamy soil with excellent drainage.
- Preferred pH: **6.0 – 7.5**.
- Avoid saline and poorly drained soils.

NUTRIENT REQUIREMENTS

- **Nitrogen (N):** Vegetative growth
- **Phosphorus (P):** Root development
- **Potassium (K):** Essential for bunch and fruit development

PROBLEMS TO MONITOR

- Banana wilt, Nematodes, Bacterial diseases
- Nutrient deficiency, Water stress, Poor drainage

ENVIRONMENTAL PARAMETERS

Parameter	Requirement
Temperature	Warm tropical conditions
Humidity	Relatively high
Sunlight	Good sunlight
Soil pH	6.0 – 7.5
Soil Type	Deep, fertile, well-drained loam

DRIP WATER QUANTITIES (TNAU STAGE-BASED)

Growth Stage	Approx. Water
After planting	4 L / plant / day
Juvenile phase	8 – 10 L / plant / day
Critical growth	12 L / plant / day
Flower bud stage	16 – 20 L / plant / day
Shooting / bunch dev.	20+ L / plant / day

AgriSpike Recommendation

- Low moisture → **Irrigation required**
- Adequate moisture → **No irrigation**
- Very high moisture → **Drainage warning**
- Low K → **Potassium deficiency warning**
- High temperature → **Heat-stress warning**

Crop Requirement Matrix Summary

Data Parameter	Tomato	Chilli	Brinjal	Onion	Banana
Temperature	✓	✓	✓	✓	✓
Soil Moisture	✓	✓	✓	✓	✓
Humidity	✓	✓	✓	✓	✓
Soil pH	✓	✓	✓	✓	✓
Nitrogen (N)	✓	✓	✓	✓	✓
Phosphorus (P)	✓	✓	✓	✓	✓
Potassium (K)	✓	✓	✓	✓	✓
Sunlight	✓	✓	✓	✓	✓
Irrigation Need	✓	✓	✓	✓	✓
Soil Type	✓	✓	✓	✓	✓
Growth Stage	✓	✓	✓	✓	✓
Pest / Disease	✓	✓	✓	✓	✓
Rainfall	✓	✓	✓	✓	✓